NFC 100'S RUPEE STUDENT'S BANK

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Declaration

We declare that the work contained in this thesis is our own, except where explicitly stated otherwise. In addition, this work has not been submitted to obtain another degree or professional qualification.

ACKNOWLEDGEMENT

In the name of Allah, whose oneness, uniqueness, and wholeness are truly unchallengeable; the Most Merciful and Compassionate. All praises to Almighty Allah. Prayers and peace be upon His last Prophet, Hazrat Muhammad (P.B.U.H.).

First and foremost, we must express our limitless gratitude to Allah, the Magnificent, for His countless blessings and assistance. We also owe a deep debt of gratitude to our university for providing us with the opportunity to complete this thesis. We would like to extend our heartfelt thanks to our highly respected project supervisor, Mr. Asim Mubarak, for his guidance, encouragement, and constructive suggestions throughout our project. Additionally, we are extremely grateful to Dr. Muhammad Sultan Zia, our head of the Computer Science department, for offering invaluable facilities, guidance, and moral support that helped us accomplish this work. Moreover, we want to express our sincere gratitude to our parents for their unwavering support throughout our lives and especially during our pursuit of our engineering degree. Thanks to their love and prayers, we were able to complete this project.

DEDICATION

We dedicate our work and efforts to Allah, the Creator of this universe, who is most Bountiful and Merciful. He is always with us, guiding us toward our destiny, as well as to the last Holy Prophet, Hazrat Muhammad (P.B.U.H), who is the reason for the creation of this universe. We also want to express our gratitude to our parents, who sacrifice for us in countless ways, and to our respected teachers, whose dedication and care have led us to this stage of knowledge.

ABSTRACT

The NFC 100's Rupee Student's Bank is a student-led fintech initiative designed to address the financial challenges faced by university students. By enabling every student to contribute a minimum of PKR 100, the platform creates a collective fund used to provide interest-free loans to peers in need. This mobile-based solution features a secure authentication system, loan application module, and transparent tracking to ensure fairness and accountability. Focused initially on NFC IEFR, the system is built with scalable technologies to support future expansion to other universities across Pakistan. The project combines ethical finance, user-friendly design, and community empowerment to promote educational continuity through peer-supported financial assistance.

List of Keywords

Keywords	Term
DFD	Data Flow diagram
ERD	Entity relationship diagram
API	Application Programming Interface

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Chapter 1: Introduction

1.1. Background:

The NFC 100's Rupee Student's Bank is a FinTech-inspired initiative created to support students facing financial difficulties in higher education. It offers a platform where students can access interest-free loans, contribute to a collective welfare fund, and manage secure, transparent transactions—all grounded in ethical, community-driven financial practices.

Over the past few decades, Financial Technology (FinTech) has significantly reshaped the global financial system. Initially, in the late 20th century, FinTech was largely focused on enhancing back-end processes within traditional banks. However, with the rise of the internet and mobile technologies in the early 2000s, it began shifting toward consumer-facing services [1]. This transition led to the development of peer-to-peer lending platforms, mobile payments, and digital wallets, all of which made financial services more accessible and user-friendly [2].

In the last ten years, the industry has grown rapidly thanks to advances in artificial intelligence, blockchain, and the widespread use of smartphones. These tools have enabled financial systems to reach underserved populations, especially in regions lacking formal banking infrastructure [3].

Inspired by this global movement, the NFC 100's Rupee Student's Bank aims to apply FinTech principles within an academic context specifically at NFC-IEFR where many students are unable to afford semester fees and often fall outside the scope of conventional bank loans or government schemes. By offering an interest-free lending model built on peer contributions and transparency, the project presents a community-first solution to student financial distress.

Ultimately, this platform reflects FinTech's broader mission: making finance more inclusive and accessible. In doing so, it empowers students to overcome financial challenges and focus fully on their academic success.

1.2. Project Overview:

The NFC 100's Rupee Student's Bank is a mobile-first platform specifically developed to serve the student body at the NFC Institute of Engineering and Fertilizer Research (NFC-IEFR). Through this application, students are able to contribute a minimum of PKR 100 each month to a shared welfare fund that supports their peers. In return, those who are unable to meet their academic expenses can apply for interest-free loans directly through the app. The platform also offers real-time visibility into the status and usage of

funds, ensuring a transparent financial environment. Loan applications submitted by students are reviewed manually by authorized university staff via a secure administrative portal, maintaining accountability and fairness in the approval process. By eliminating the burden of interest and simplifying the process of accessing financial aid, the NFC 100's Rupee Student's Bank fosters a culture of trust, transparency, and collective responsibility within the student community.

1.3. Objectives:

The primary objective of this project is to create a transparent, ethical, and student-driven financial aid platform, the platform aims to promote collective responsibility and financial empowerment within the academic community. The key objectives include:

a) Promote Financial Inclusion: Provide interest-free loans to students who are unable to afford tuition and other academic expenses, offering a practical alternative to conventional banking systems that often exclude low-income students.

b) Encourage Student Participation: Create a culture of mutual support by enabling students to contribute small monthly amounts to a shared welfare fund that benefits their peers.

c) Ensure Transparency: Offer real-time visibility into the flow of funds, including donations and loan disbursements, so that all students donors, recipients, and administrators can track transactions with full accountability.

d) Reduce Financial Stress: Prevent educational interruptions by offering timely financial relief, helping students remain enrolled and focused on their academic goals.

e) Replicable Model: Develop a scalable system that can serve as a framework for other universities to implement student-led, interest-free financial support initiatives.

1.4. Scope of the Project:

The scope of this project encompasses the design, development, and implementation of a mobile-based platform that enables NFC-IEFR students to contribute to a shared welfare fund and apply for interest-free loans. The system includes a student-facing mobile application for registration, donations, loan applications, and view analytics as well as a web-based admin portal for managing and verifying, users, donation amount and loan requests.

The application is intended to serve only the NFC-IEFR student community at this stage, its infrastructure is designed to support future enhancements and broader institutional adoption.

1.5. Significance and Benefits:

The NFC 100's Rupee Student's Bank holds substantial value for the student community at NFC-IEFR. It not only offers practical financial assistance but also introduces a sustainable, student-powered support model. Key benefits include:

a) Convenience for Students: Students can access financial aid without the burden of bank procedures or interest rates.

b) Empowerment: The initiative fosters a sense of responsibility, giving students ownership over their financial ecosystem.

c) Transparency: Every donation and loan are traceable, ensuring trust among students that how funds are collected, allocated, and used.

d) Academic Continuity: Timely loan access can prevent students from dropping out due to temporary financial setbacks.

e) Replicable Impact: This model can serve as a blueprint for student welfare systems in other universities, particularly where existing state or private financial support mechanisms fall short.

Chapter 2: Motivations and Related Literature

2.1. Motivation for the Project:

The NFC 100's Rupee Student's Bank is driven by a vision to empower students through ethical, accessible, and innovative financial solutions. The motivation behind this project emerged through research, inspiration from established ethical models, and practical constraints experienced while attempting to launch a full-scale digital bank.

2.1.1. Inspiration From Akhuwat Foundation:

This project draws inspiration from the Akhuwat Foundation [4], which has demonstrated the transformative potential of interest-free banking in fostering equity, social empowerment, and financial inclusion. Akhuwat Foundation's pioneering model of providing interest-free microfinance (Qarz-e-Hasna) has disbursed over 4.5 million loans, amounting to PKR 128 billion (US$798 million), to more than 3 million families across Pakistan. This approach has empowered marginalized communities to overcome financial barriers and become active societal contributors. Inspired by this model, we sought to extend similar principles into the academic sphere by creating a platform exclusively for students.

2.1.2. The Original Vision "100's Bank":

Our journey began with an ambitious idea to establish a comprehensive digital bank, initially named "100's Bank." The goal was to offer interest-free loans alongside modern financial services, including microfinance, green financing, bill-splitting, buy-now-pay-later options, and customizable credit cards. This vision was conceived as a response to the absence of digital banking solutions in Pakistan that align with ethical financial principles and cater specifically to underserved segments, such as a modern financial platform that provides interest-free loans while adhering to Islamic principles.

2.1.3. Regulatory and Financial Challenges:

However, through extensive research, we discovered that making this vision a reality would require compliance with several regulatory frameworks, including:

1. **State Bank of Pakistan (SBP)** [5]**:** For licensing, compliance with digital banking frameworks, and capital adequacy requirements. SBP requires digital banks to maintain a minimum paid-up capital of PKR 10 billion.

2. **Securities and Exchange Commission of Pakistan (SECP)** [6]**:** For incorporation, regulatory oversight, and financial governance.

3. **Federal Board of Revenue (FBR)** [7]**:** For taxation matters, including income tax, GST registration, and withholding tax compliance.

4. **Pakistan Telecommunication Authority (PTA)** [8]**:** For acquiring permissions related to SMS gateway services, digital communication protocols, and mobile banking networks.

5. **National Database and Registration Authority (NADRA)** [9]**:** For biometric identity verification, user KYC (Know Your Customer) compliance, and digital onboarding mechanisms.

In addition to the stringent regulatory requirements, we calculated that a budget of PKR 10,000 million would be necessary to fulfill compliance obligations and develop the infrastructure for a digital bank. Following consultations with senior bankers and tech industry experts, we concluded that launching a digital bank at this stage would be capital-intensive and time-consuming, requiring at least two to three years for successful implementation.

2.1.4. Shift to NFC 100's Rupee Student's Bank:

Recognizing these constraints, we refined our approach to focus on our core mission—offering interest-free loans. This led to the conceptualization of the NFC 100's Rupee Student's Bank, a streamlined, student-centered initiative designed to solve financial problems faced by students while laying the groundwork for future expansion into a digital banking platform. By adopting Akhuwat Foundation's [4] ethical lending principles and focusing on a single impactful service, we aim to create a solution that is transparent, accessible, and sustainable.

The NFC 100's Rupee Student's Bank represents our first step toward bridging financial gaps for students in need, fostering a culture of mutual support and inclusivity within academic communities. Inspired by Akhuwat Foundation's [4] legacy and our original vision for 100's Bank, we aspire to grow this initiative into a robust, digitally driven platform in the future, ensuring that students have the tools they need to thrive without the burden of interest-based debt.

2.2. Literature Review:

2.2.1. Background Research:

The design and development of the NFC 100's Rupee Student's Bank stem from a thorough exploration of existing financial models, particularly those implemented by traditional banking systems and welfare trusts. By examining these frameworks, we identified critical insights and strategies that informed the creation of an ethical, inclusive financial platform tailored to address the unique challenges faced by students.

1. Traditional Banks:

- **National Bank of Pakistan (NBP) Student Loan Scheme**:

The NBP Student Loan Scheme [10] provides interest-free loans to students pursuing degrees in science, engineering, and technical fields. This government-backed initiative highlights the importance of thorough needs verification and transparent fund distribution, both vital aspects of our project. However, a significant limitation is that our university is not listed among the recognized institutions under this scheme, which means many deserving students cannot access its benefits. This gap emphasizes the necessity for a more inclusive financial solution that directly supports students from various educational backgrounds.

- **Habib Bank Limited (HBL)**:

The HBL Student Loan Program [11] provides financial assistance to students; however, it operates on an interest-bearing model with a high interest rate of approximately 35%, unlike government-backed initiatives. While this program aims to help students manage their educational expenses, the significant interest rate creates a substantial financial burden, making it less accessible for those from low-income backgrounds. In contrast, NFC 100's Rupee Student's Bank focuses on offering ethical, interest-free loans to reduce financial barriers. The interest-based structure of HBL highlights the need for alternative solutions, such as NFC 100's Rupee Student's Bank, to provide inclusive and sustainable financial support tailored to the needs of students.

2. Welfare Trusts:

- **Akhuwat Foundation**:

The Akhuwat Foundation [4] offers interest-free educational loans ranging from Rs. 10,000 to Rs. 50,000 for students pursuing post-matriculation to master's degrees. These loans have a repayment period of 10 to 24 months and require an application fee of up to Rs. 200. While this initiative provides critical financial support, the limited loan amount often falls short of meeting the increasing costs of education, making it difficult for students to continue their studies effectively. To address this limitation, the NFC 100's Rupee Student's Bank aims to provide a more inclusive and tailored solution. By focusing on the specific needs of our students and offering flexible, interest-free loans, this platform seeks to bridge the financial gap and ensure students can pursue their education without unnecessary barriers.

- **Ihsan Trust**:

Ihsan Trust's [12] ethical lending model, which provides interest-free loans (Qarz-e-Hasna) to financially constrained individuals, has significantly influenced our focus on ethical and inclusive financial solutions. This trust collaborates with various universities across Pakistan to support students, ensuring that financial barriers do not hinder their education. However, it was observed that our university does not have a formal collaboration with Ihsan Trust, limiting access to this valuable resource for many deserving students. This gap underscores the need for an initiative like NFC 100's Rupee Student's Bank, specifically designed to address the financial challenges faced by students at our institution.

- **University Welfare Trusts (NUST and LUMS)**:

➢ **NUST Trust Fund [13]**: Collaborates with the Ihsan Trust Foundation to offer interest-free loans. NUST collects application data and forwards it to Ihsan Trust, which funds deserving students.

➢ **LUMS Student Personal Loan Product [14]**: Also partners with Ihsan Trust, employing rigorous verification processes to ensure that only eligible students receive financial aid.

The operational models of these university welfare trusts demonstrated the value of partnerships and transparent fund distribution mechanisms. However, they also highlighted the need for a platform that is independent of external collaborations and directly accessible to students.

2.2.2. Key Findings:

The comparative analysis of traditional banks and welfare trusts revealed several critical insights that shaped the NFC 100's Rupee Student's Bank:

a) Ethical Lending: Models like Akhuwat Foundation [4] and Ihsan Trust [12] emphasize the feasibility of providing interest-free loans while maintaining financial sustainability.

b) Accessibility: The limitations of traditional bank loans, such as high interest rates and restrictive eligibility criteria, underscore the importance of creating an inclusive, student-centric solution.

c) Collaboration vs. Independence: While collaborations can enhance credibility and sustainability, the absence of formal partnerships with our university highlighted the need for an independent financial platform dedicated to student welfare.

d) Transparency and Verification: Rigorous application and verification processes, as demonstrated by traditional banks and welfare trust models, are essential for ensuring fair and effective loan distribution.

Based on the review of traditional banks and welfare trust-based models, the NFC 100's Rupee Student's Bank has been conceptualized as an interest-free loan platform specifically designed for students. This initiative aims to address the immediate financial challenges faced by students while operating within our current economic and operational constraints. By starting small, we can establish a solid foundation that emphasizes transparency, accountability, and community engagement. As the project grows and we secure sufficient capital, we plan to expand the NFC 100's Rupee Student's Bank into a comprehensive digital bank that aligns with our original vision.

2.3 Identifying the Problem:

“There is a lack of financial support, welfare funds, or scholarship programs specifically available to students of NFC-IEFR, which limits their access to essential resources for academic and personal development.”

This pressing issue results in educational disruption, financial stress, and limited opportunities for deserving students. Traditional financial aid programs often exclude NFC-IEFR or require partnerships that the institution does not have.

Chapter 3: Proposed Approach

3.1 Platform Functionality Overview:

3.3.1 User Interface and Experience

The proposed approach for the NFC 100's Rupee Student's Bank centers on the development of a student-focused, mobile-based platform that facilitates financial assistance through interest-free, peer-supported loans. The system is designed to offer a seamless experience for both students in need and the university staff managing the process. The platform aims to promote financial inclusion, foster a culture of mutual aid among students, and ensure transparency in fund collection and distribution.

Functionally, the platform allows students to register, contribute monthly donations to a shared welfare fund, and apply for loans directly through a secure mobile application. Meanwhile, the administrative web portal enables authorized university staff to authenticate student records, review loan applications, and manage the disbursement process. The system ensures that all financial activity—donations, loan submissions, approvals, and analytics—is securely recorded and accessible in real time. By combining accessibility, trust, and transparency, the NFC 100's Rupee Student's Bank creates an inclusive financial support structure within the academic environment.

3.2 System Design:

3.2.1 Flowchart:

The Flowchart showcases the step-by-step process of various user interactions on the platform, outlining the sequence of actions and decision points.

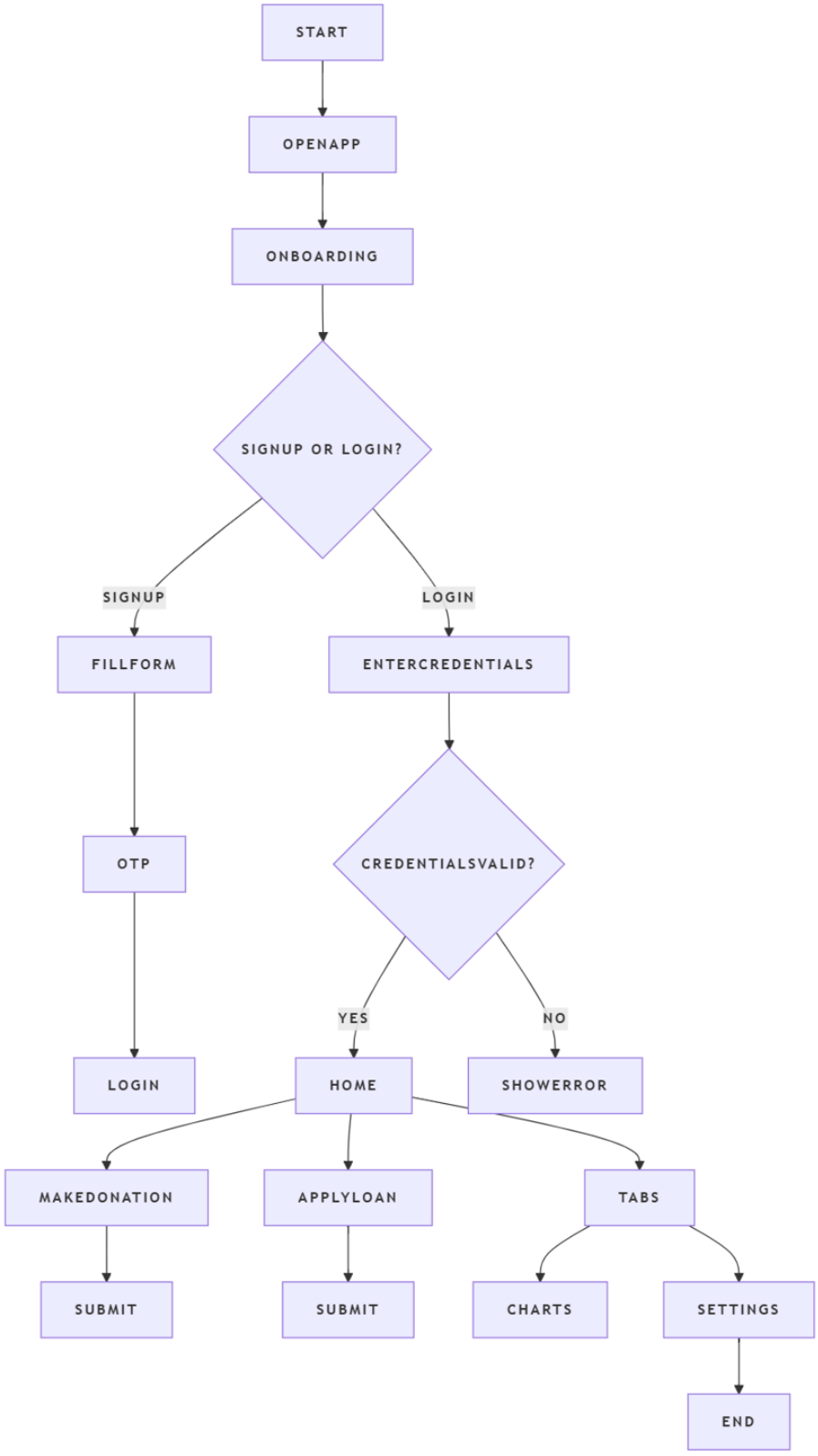


Figure 1: Flowchart for the Student Mobile App

The above Figure shows that student flow begins with the app launch, displaying a splash screen followed by an onboarding screen where the user chooses to either sign up or log in. If the user is new, they complete the sign-up form and proceed through OTP verification before accessing the login page. After entering valid credentials, they are directed to the home screen.

From the home screen, students can either make a donation or apply for a loan. The donation process involves entering payment details and submitting the form. The loan application process consists of filling out personal, family, and additional information in multiple steps, followed by submission.

Navigation options at the bottom of the screen allow users to view donation analytics (charts) and manage their profile through settings. The flow ends once the user exits the app or logs out.

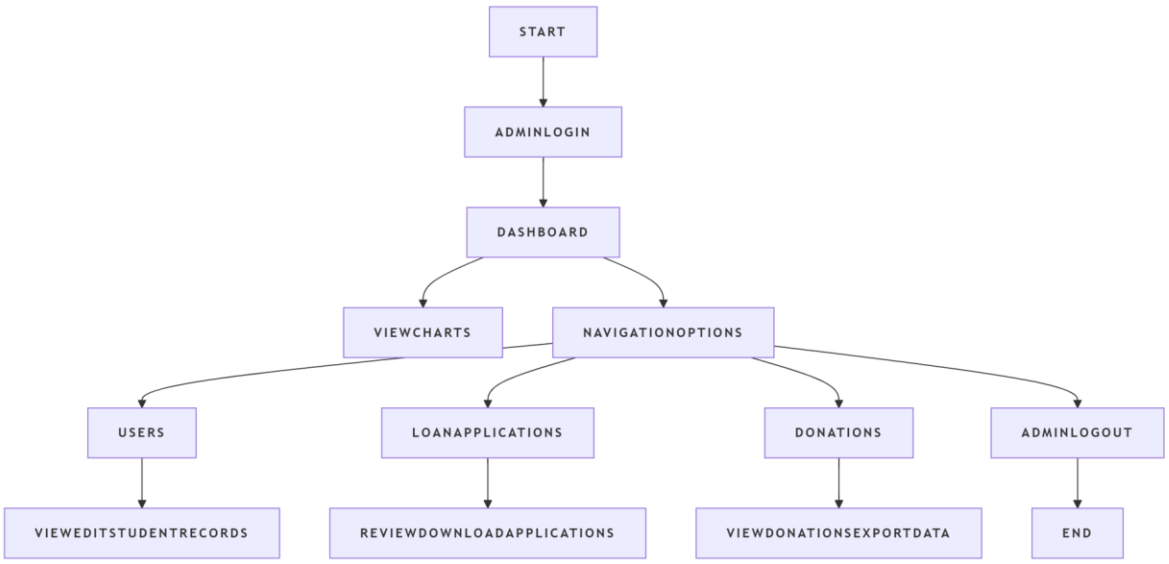


Figure 2: Flowchart for Admin Web Portal

The above Figure shows that admin flow begins with a secure login to the web portal. After logging in, the admin accesses a dashboard displaying key statistics about users, loan applications, and donations.

The admin can navigate through different sections: view and manage student user data, review submitted loan applications, and monitor donation activity. The admin also has the ability to export relevant data, such as donation records and phone numbers for communication purposes. The flow concludes when the admin logs out of the portal.

3.2.2 Context Level Data Flow Diagram (DFD):

The Context Level DFD illustrates the interactions between the platform and external entities. It provides a high-level overview of how data flows into and out of the system, identifying the main processes involved in the NFC 100's Rupee Student's Bank.

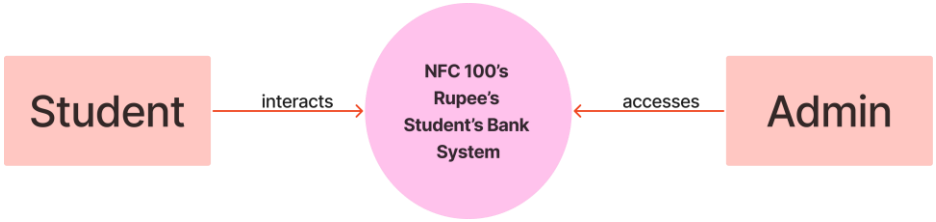


Figure 3: Context Level Data Flow Diagram (DFD)

The above Figure represents the entire NFC 100's Rupee Student's Bank system as a single process that interacts with two main external entities: Student and Admin. Students interact with the system to sign up, log in, apply for loans, make donations, and view analytics. Admins access student data, review loan applications, and monitor donations and analytics.

The system exchanges data with external entities and stores it in internal data stores, such as Student Data, Loan Applications, Donations, Analytics, and Documents.

3.2.3 DFD1 and DFD2:

DFD (1)

DFD1 and DFD2 are detailed Data Flow Diagrams that expand on the processes identified in the Context Level DFD. DFD1 breaks down the major processes into sub-processes, while DFD2 presents a more detailed view of data flow within the system.

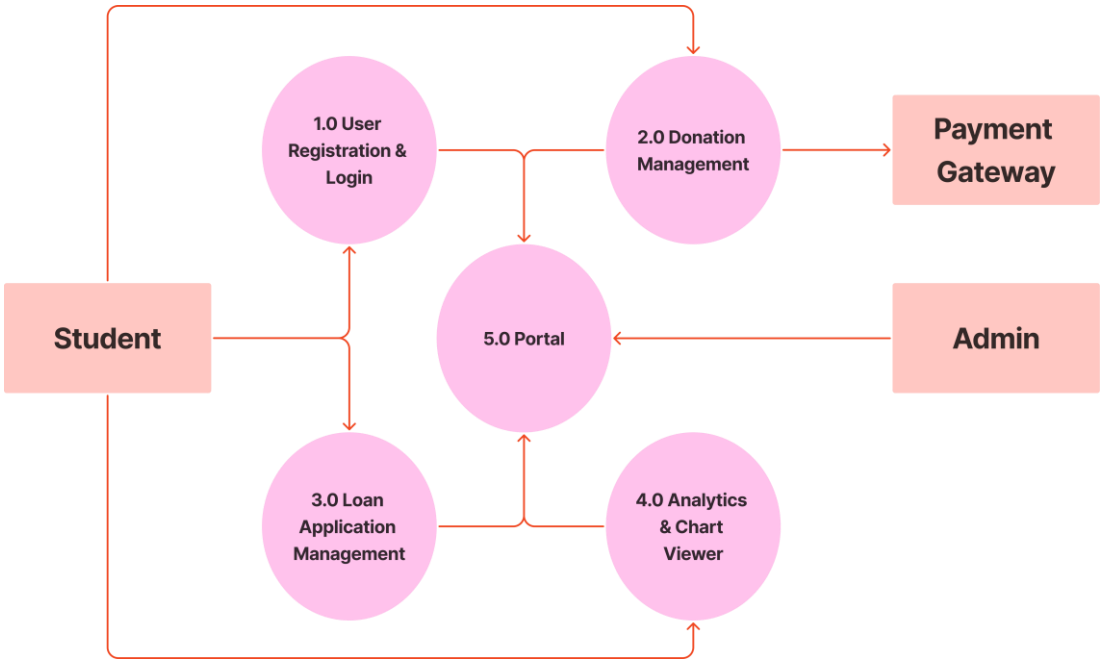


Figure 4: Data Flow Diagram (1)

The above DFD Level 1 presents a simplified, high-level view of the system, focusing on the main processes and their interactions with external entities. It illustrates how students and admins interact with five core processes: registration and login, donation management, loan application management, analytics viewing, and the central portal. The student initiates actions like registering, donating, applying for loans, and viewing charts, while the admin oversees and manages these activities through the portal. Donations are processed via the payment gateway.

DFD (2)

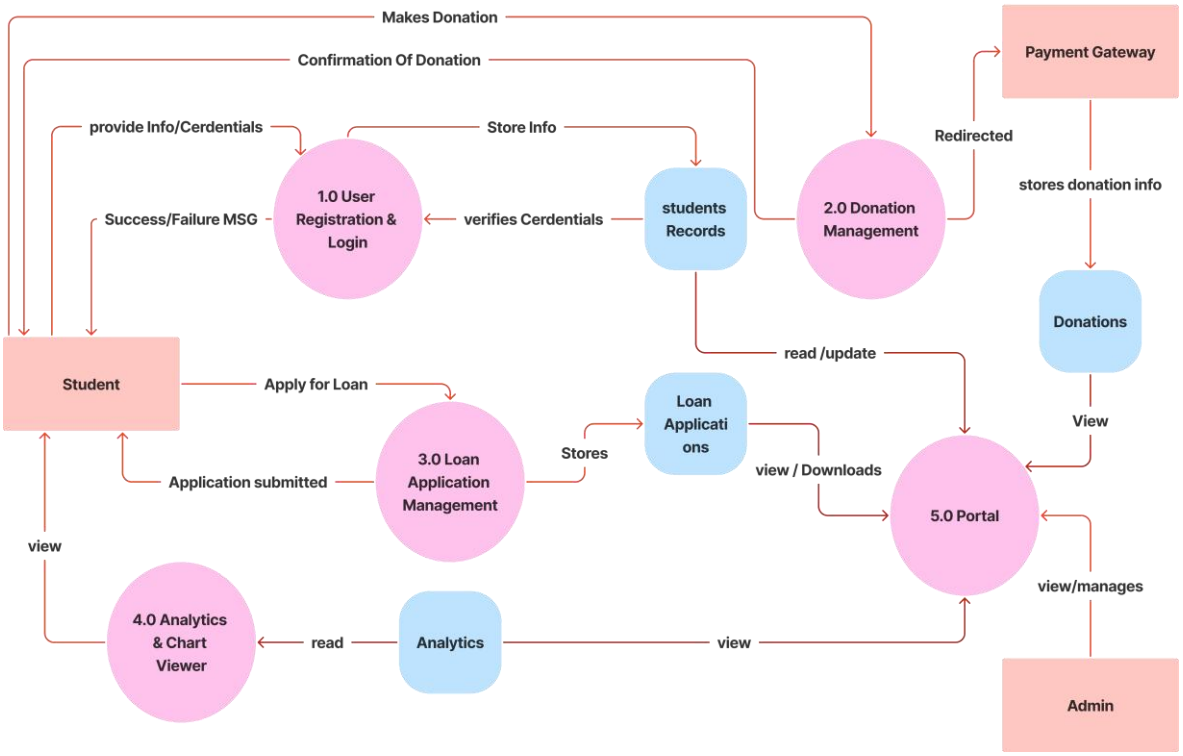


Figure 5: Data Flow Diagram (2)

In the Data Flow diagram (DFD) Level 2 diagram breaks down the core processes of the system and introduces internal data stores. Students first interact with the User Registration & Login process, storing their data in the Student Records. They can make donations via the Donation Management process, which uses the Payment Gateway and stores results in the Donations data store. Loan requests are handled through the Loan Application Management process and saved in the Loan Applications store. Students and admins can view visual insights through the Analytics & Chart Viewer, which pulls data from the Analytics store. The Portal process acts as the central hub, connecting all other modules. Admins access the portal to manage users, donations, loans, and analytics. All actions flow through the Portal for centralized control. Data is stored and retrieved from relevant stores for each process. The diagram highlights secure, organized interaction between users, system processes, and external services.

3.2.4 Entity-Relationship Diagram (ERD):

The Entity-Relationship Diagram (ERD) illustrates the structured data model of the NFC 100's Rupee Student's Bank system. It defines the key entities involved, their attributes, and how they relate to each other.

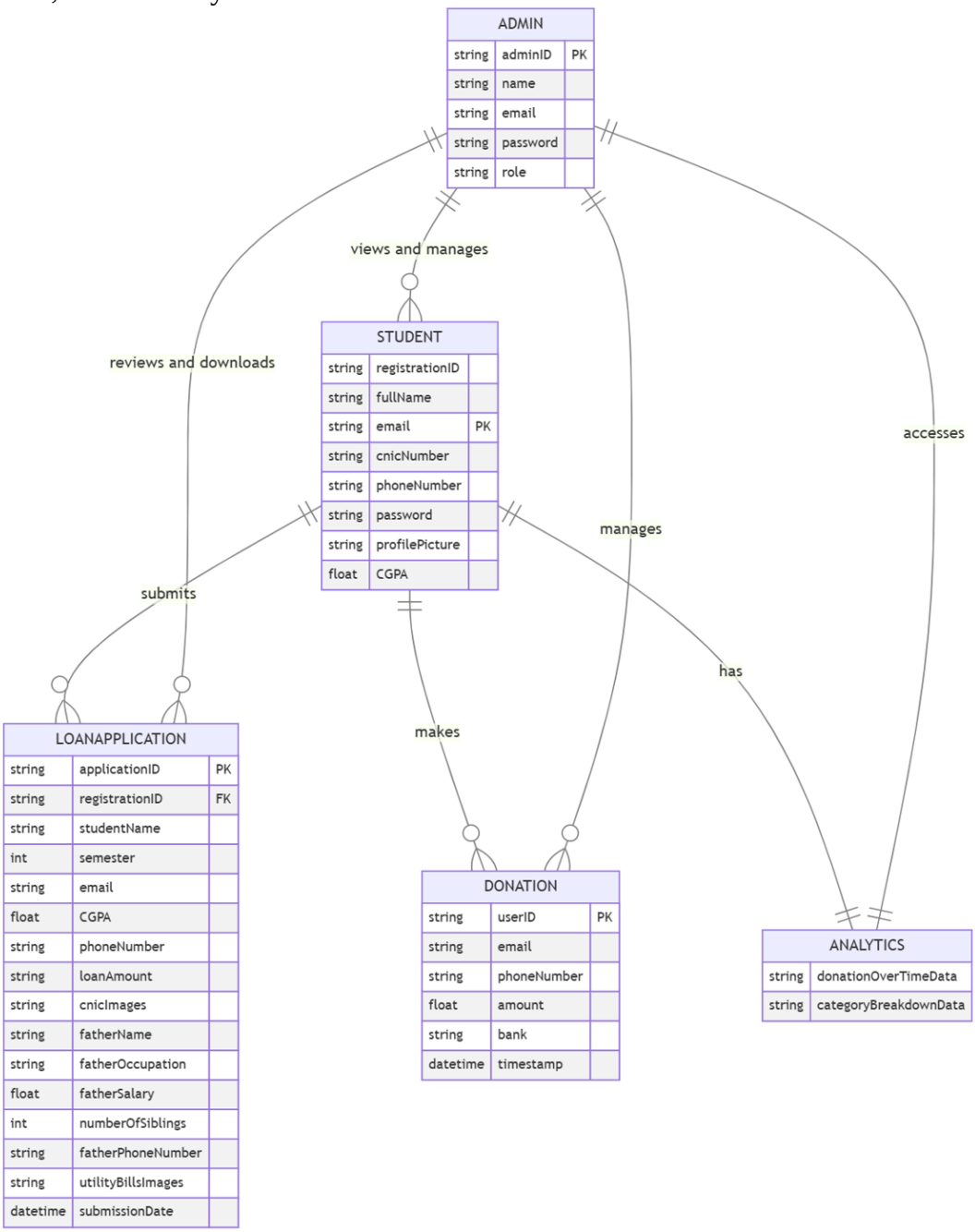


Figure 6: Entity-Relationship Diagram (ERD)

- **Student and Donation**: There is a relationship between students and donations, as students contribute funds through the donation feature. This is a **one-to-many** relationship, where one student can make multiple donations, but each donation is linked to a single student.

- **Student and Loan Application**: Students apply for loan applications. This forms a **one-to-many** relationship, where a student can submit multiple loan applications, but each application is submitted by only one student.

- **Student and Analytics**: Analytics tracks a student's donation and loan history. This is a **one-to-one** relationship, where each student has one analytics record, and each analytics record belongs to only one student.

- **Admin and Student**: Admins manage all student users in the system. This forms a **one-to-many** relationship, where one admin can manage multiple students, but each student is managed by only one admin (in the context of system control).

- **Admin and Loan Application**: Admins review and process loan applications submitted by students. This is a **one-to-many** relationship, where one admin can handle many loan applications, but each loan application is reviewed by a single admin.

- **Admin and Donation**: Admins oversee and manage student donations. This is a **one-to-many** relationship, where one admin can monitor many donations, while each donation is reviewed or visible to one admin.

- **Student and Admin:** Admins manage multiple students through the web portal, viewing and updating their data. This creates a **one-to-many** relationship where one admin manages many students, but each student is managed by only one admin at a time.

3.2.5 Use Case Diagram:

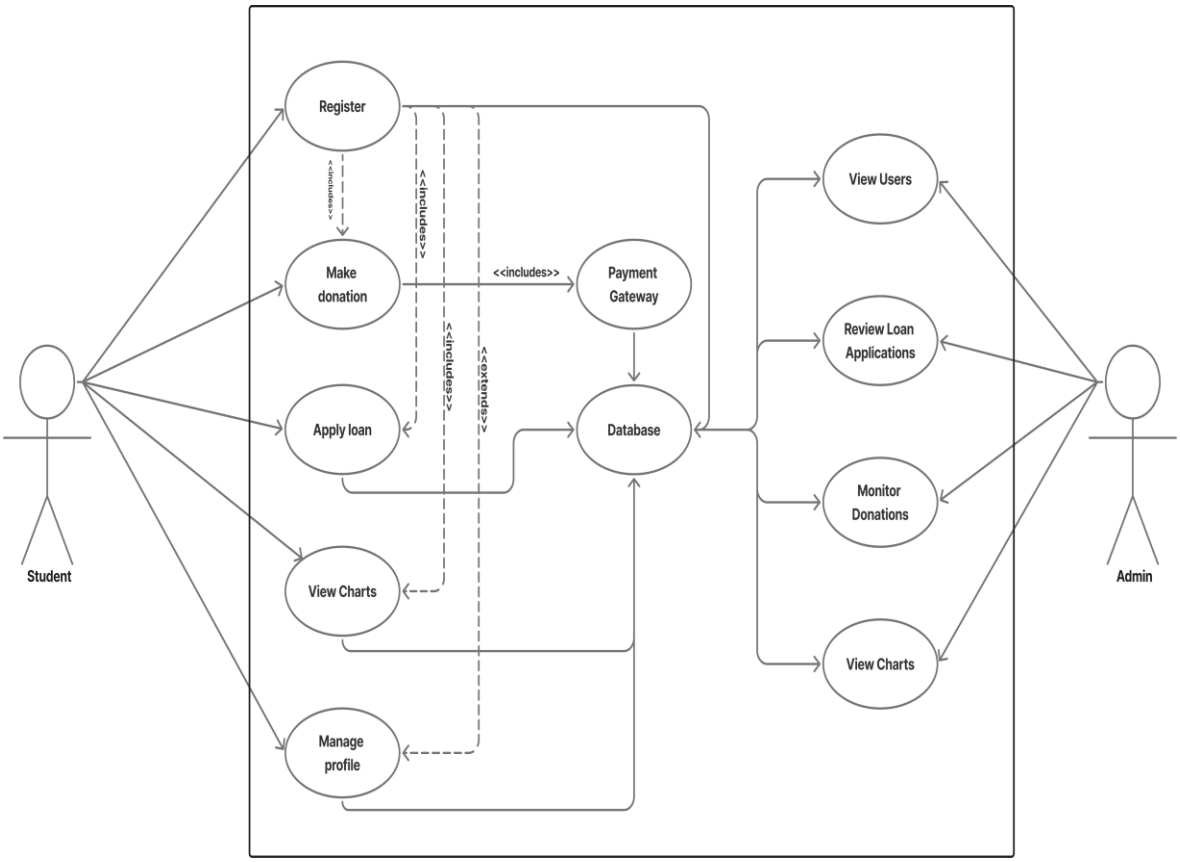


Figure 7: Use Case Diagram

In the Use Case Diagram there are two primary actors: Student and Admin, each with distinct roles and responsibilities within the application.

The student actor initiates the process by registering in the app. Registration acts as the gateway to other core functionalities such as making donations, applying for loans, viewing donation and loan charts, and managing their profile. The registration use case connects logically to these other use cases, indicating that only registered students can perform these actions. Students can make donations, which involve an interaction with the Payment Gateway before the transaction data is stored in the Database.

The admin actor interacts primarily with the backend system through the web portal. Admins can view and manage all users (students), monitor donation records, review loan applications, and analyze charts representing overall app performance. The admin's

actions are directly connected to the Database, showing that their management and monitoring functions update and retrieve data from the central storage.

The Payment Gateway serves as an intermediary between the student's donation action and the database, ensuring secure processing of financial transactions. Donations flow from the student to the payment gateway and then to the database.

The Database is centrally placed in the diagram, acting as the hub where all key data is stored, accessed, and updated. Arrows show data flowing in from the student's use cases except makes donations, which route through the payment gateway and out to the admin's monitoring and management use cases.

This structure reflects the real-world flow of data and responsibilities, with students engaging the app features after registration, donations processed securely via payment, and admins maintaining oversight through database interactions.

3.3 Platform Design and Features:

The platform's design and features are strategically planned to cater to the needs of both Users and Admin. The following subsections elaborate on the essential components of the platform

3.3.1 User Interface and Experience:

The app is designed with a clean and intuitive UI to cater to the needs of student users. It features minimal navigation, quick access to donation and loan options, and a personalized dashboard that displays donation totals, loan status, and financial insights. The onboarding process is simple, ensuring accessibility even for first-time users.

3.3.2 Student Registration and Verification:

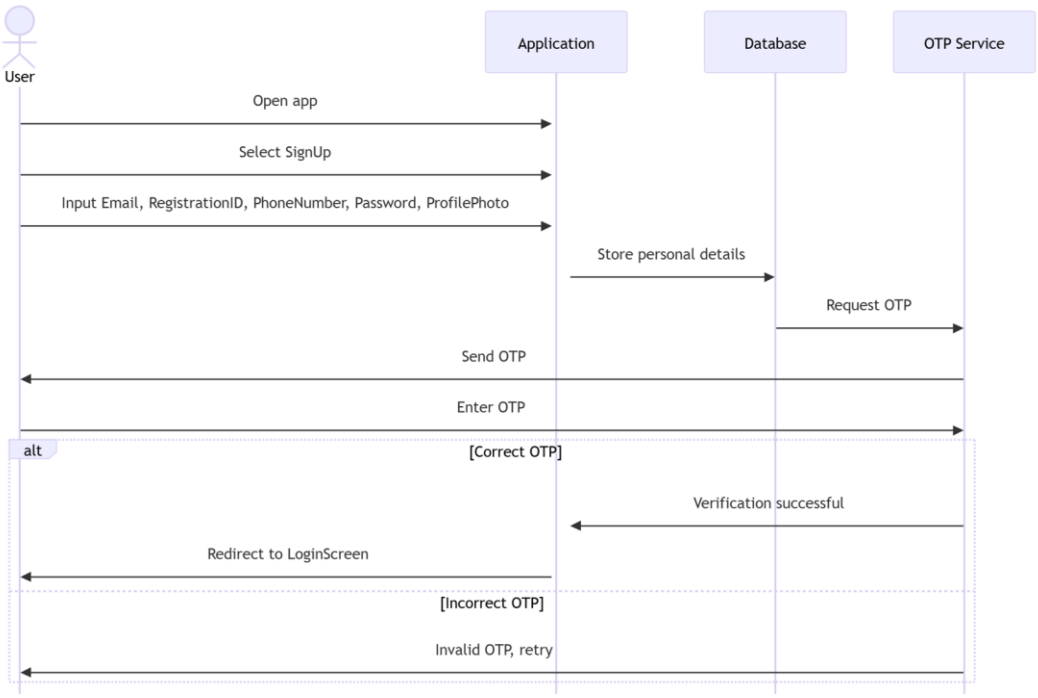


Figure 8: Student Registration Sequence Diagram

The registration and verification sequence begins when a student opens the mobile application and navigates to the Sign-Up screen. Here, the student enters personal details including their email, registration ID, phone number, password, and profile picture. Upon submission, this data is sent to the server and securely stored in the database. An OTP is then generated and sent to the student for identity verification. After the student enters the OTP, the system checks for correctness. If verified successfully, the student is redirected to the login screen, completing the registration flow securely.

3.3.3 Student Login and Dashboard Access:

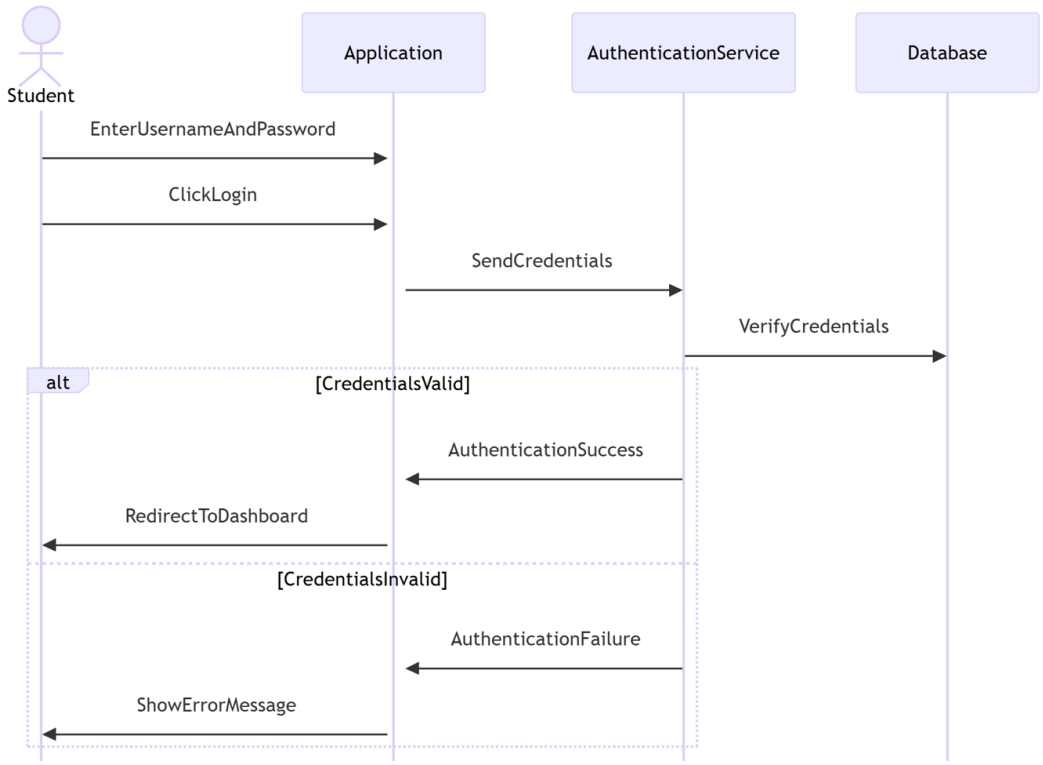


Figure 9: Student Login Sequence Diagram

Once registered, the student accesses the app by entering their login credentials email and password. The login request is sent to the backend server, which then verifies the credentials against existing records in the database. If the information matches, the system grants access to the home screen or dashboard, where core features like donation, loan application, and analytics are available. If the credentials are invalid, an error message is returned and shown on the screen, prompting the user to try again or reset their password.

3.3.4 Donation Workflow:

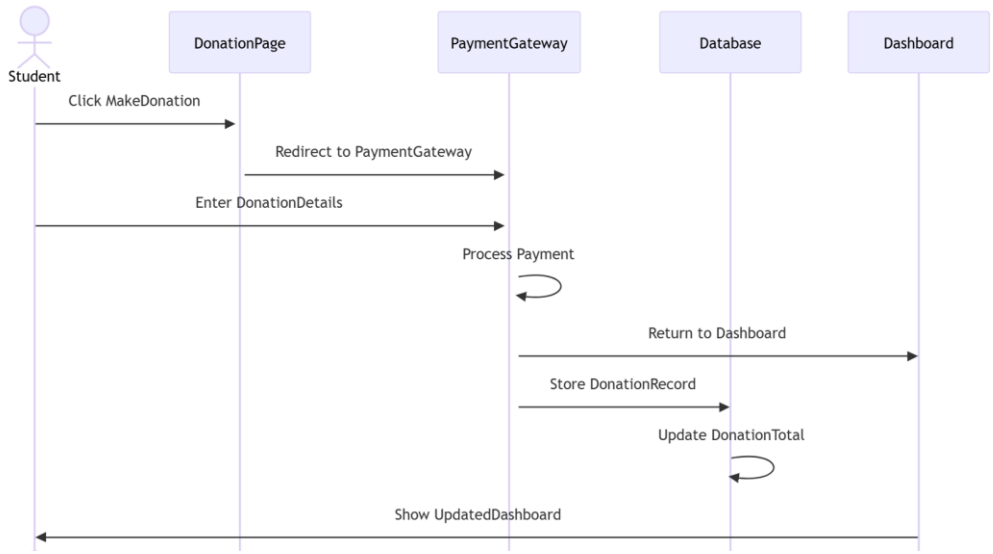


Figure 10: Donation Sequence Diagram

In this sequence, a logged-in student chooses the "Make a Donation" option from the dashboard. The app redirects the student to a payment gateway where they input details such as their email, phone number, donation amount, and selected payment method or bank. Upon successful transaction, the system records the donation in the database and updates the total amount donated on the student's dashboard. This ensures that contributions are immediately reflected and logged for future analytics.

3.3.5 Loan Application Submission:

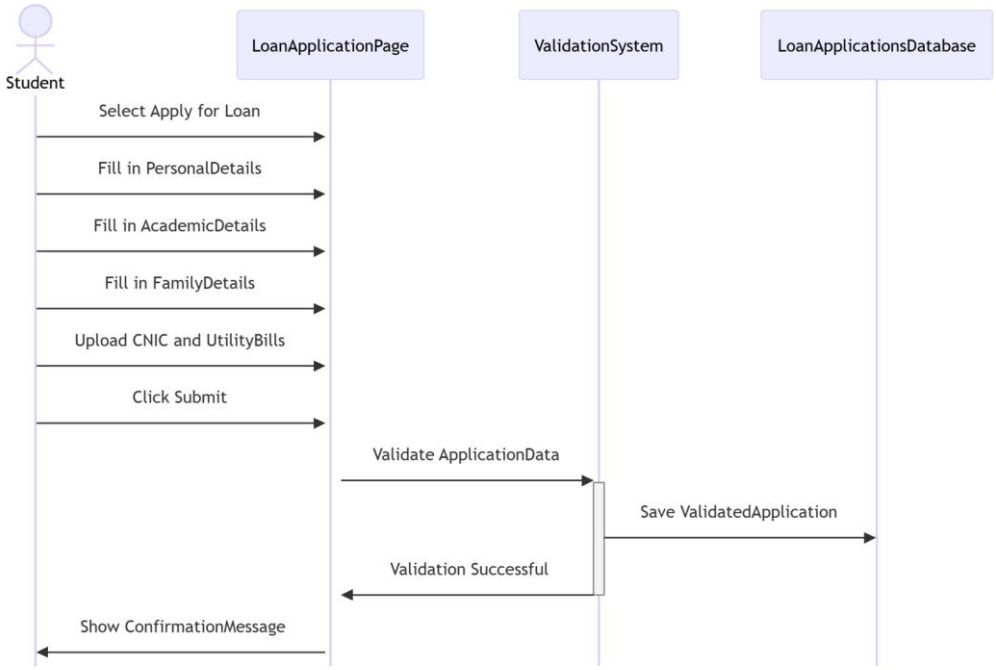


Figure 11: Loan Application Sequence Diagram

When a student decides to apply for financial assistance, they select the “Apply for Loan” feature from the dashboard. The app takes them through a step-by-step form where they provide personal, academic, and family financial information. They also upload required documents such as CNIC images and utility bills. Once all data is submitted, the backend validates the inputs and stores the loan application in the database. A confirmation message is shown to the student, notifying them that their application is under review.

3.3.6 Admin Donation and User Management:

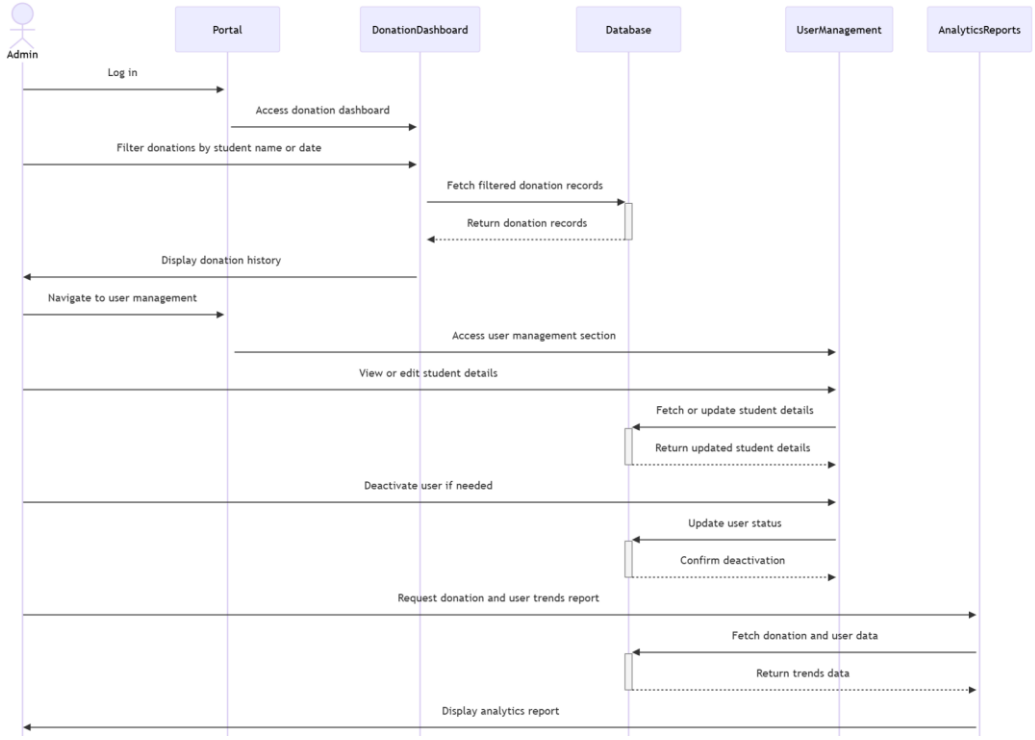


Figure 12: Admin Portal Management Sequence Diagram

This Figure illustrates the admin's interaction with the donation and user management modules through the web-based portal. After logging in, the admin accesses the donation dashboard and uses filters—like student name or date range—to retrieve relevant donation data from the database. The system displays this data for analysis. The admin can then switch to the user management section to view or edit student accounts, deactivate users if required, and generate analytical reports on student activity and donation trends. This sequence supports platform maintenance and transparency on the backend.

3.3.7 Admin Loan Management:

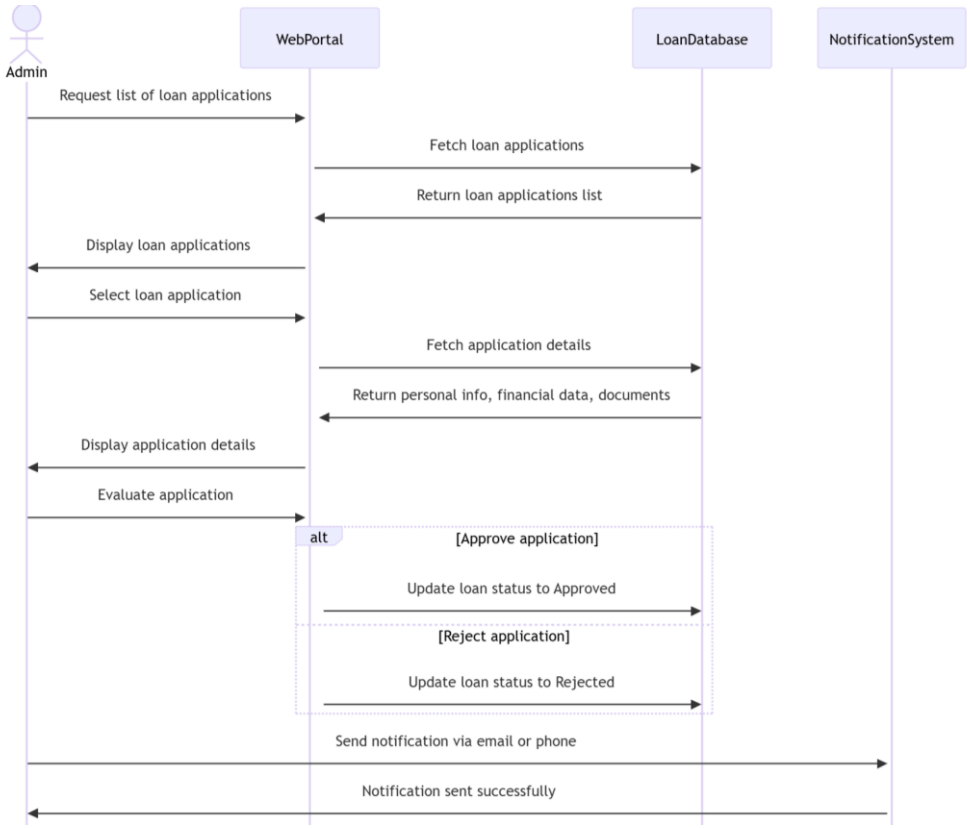


Figure 13: Admin Loan Application Management Sequence Diagram

This process begins when the admin logs into the portal and views the list of loan applications submitted by students. Selecting a specific application, the admin reviews all attached personal details and documents. After evaluation, the admin decides to either approve or reject the request. The system updates the loan status in the database accordingly. Once the decision is recorded, the admin manually notifies the student of the outcome through their registered email address or phone number. This manual notification ensures that the communication is handled with care and sensitivity.

3.4 Development Technologies and Tools:

The successful implementation of the NFC 100's Rupee Student's Bank platform depends on the integration of reliable, scalable, and secure technologies. The development stack is selected to support a smooth user experience, efficient data processing, and compatibility with Pakistan's local payment infrastructure. The following subsections outline the major components of the system architecture:

3.4.1 Frontend Technologies:

The frontend of the platform is developed using React Native, a modern, cross-platform framework ideal for building mobile applications on both Android and iOS. This choice ensures a unified codebase and consistent user interface across devices, allowing students to access donation and loan features with minimal technical barriers.

3.4.2 Backend Technologies:

The backend services are built using Node.js with Express.js, providing a lightweight and asynchronous runtime environment well-suited for real-time data operations. This technology stack supports secure authentication, API handling, and scalable performance for managing user activity, loan workflows, and admin operations.

3.4.3 Database Selection and Management:

The platform uses MongoDB, a flexible NoSQL database, to store and manage dynamic data such as student records, loan applications, donation logs, and admin actions. MongoDB allows for easy handling of complex data structures, supports horizontal scaling, and ensures high availability and performance across all operations.

3.4.4 Payment Gateway Integration:

To process financial transactions, the application integrates with JazzCash and EasyPaisa, two widely used mobile wallet services in Pakistan. These integrations allow secure, localized handling of donations through trusted third-party payment gateways, enhancing accessibility for students.

3.4.5 Cloud Hosting and Deployment:

The platform uses Docker for containerized deployment, ensuring consistent performance across environments. These containers are hosted on cloud services such as AWS or GCP, providing scalable infrastructure, secure storage, and automated backups to support reliable and maintainable system operations.

Chapter 4: Implementation

4.1 Development Environment Setup:

The implementation phase of the NFC 100's Rupee Student's Bank project began with setting up a collaborative and scalable development environment to support both frontend and backend tasks. The team prepared the essential toolchains and frameworks to streamline development and ensure code reliability. The setup prioritized maintainability, portability, and smooth team collaboration.

4.1.1 Software and Tools Configuration:

To begin development, the team installed and configured essential tools and environments. Visual Studio Code (VS Code) was used as the primary Integrated Development Environment (IDE) for writing, debugging, and managing code. For API testing and debugging, Insomnia was used, offering a clean interface for sending requests and analyzing server responses. MongoDB Compass facilitated database management, allowing the team to visualize collections and documents. Additionally, Docker Desktop was configured to support containerized development, ensuring consistency across environments and simplifying deployment.

4.1.2 Version Control and Collaboration Mechanisms

For version control and team coordination, Git and GitHub were used. A centralized GitHub repository was created to host the project code. Developers used separate branches for individual features and merged them into the main branch after testing and review. This version control system ensured that progress could be tracked, code conflicts could be resolved efficiently, and team members could collaborate seamlessly across different modules.

4.2 Database design and Implement:

A well-structured database was essential to handle various types of dynamic data within the platform, including student records, loan applications, donation logs, admin actions, and analytics. MongoDB, a flexible and document-oriented NoSQL database, was chosen for its ability to efficiently manage nested and semi-structured data. The database was designed to maintain data consistency, support real-time updates, and ensure secure

storage of user-submitted information such as personal details, utility bills, and loan history. Collections and document schemas were defined to align with the application's modular structure, optimizing both read and write operations.

4.3 Frontend Development:

Frontend development focused on building a responsive, intuitive mobile interface for students using React Native. The design aimed to provide a smooth user experience across Android and iOS devices, enabling users to navigate the app with ease. The core frontend modules included the onboarding screen, registration and login forms, donation module, loan application form, and dashboard with real-time data visualizations. The frontend was integrated with the backend APIs to allow real-time data fetching, form submissions, and status tracking. Care was taken to maintain UI consistency, accessibility, and responsiveness throughout the app.

4.3.1 UI/UX design:

These designs of different screens of app shows how the app look.

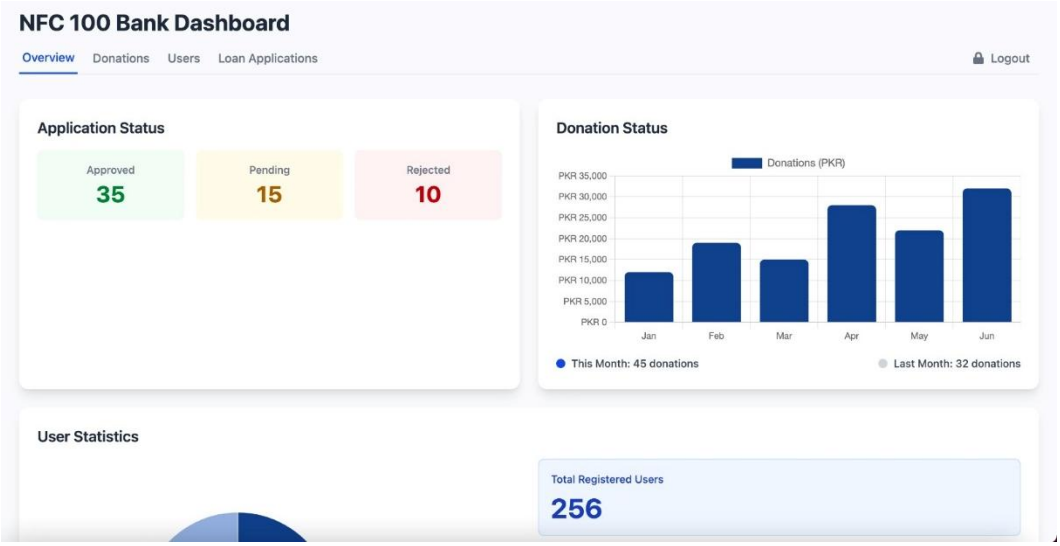


Figure 14: Admin Portal

When the user opens the app, they are first greeted with a splash screen for a few seconds. Following this, the onboarding screen appears, presenting two options: "Sign Up" and "Log In."



Figure 15: Onboarding Screen

Since the user is new, they will click on "Sign Up" and be directed to the sign-up screen. Here, they need to fill in the following information: email, full name, phone number, registration ID, password, and upload a profile picture. This information will be securely stored in the database.

After completing the sign-up form and clicking on "Sign Up," the user will be taken to the OTP verification screen. If they enter the correct OTP, they can click "Verify" and will be directed to the login screen. There is also an option to resend the OTP if needed.

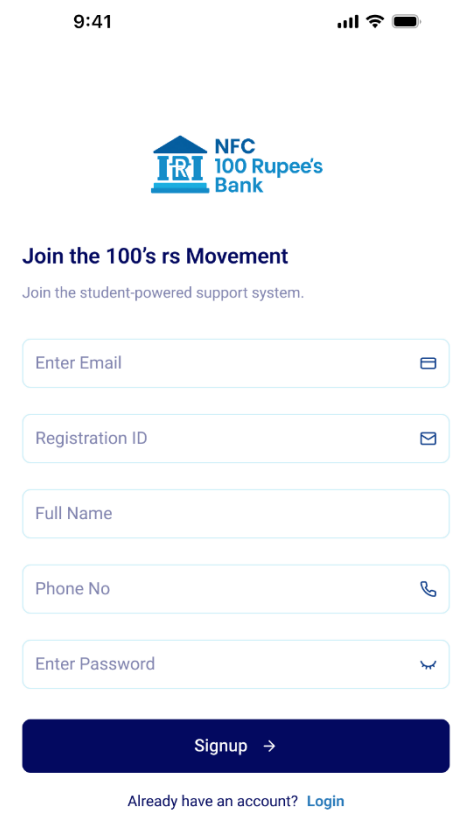


Figure 16: Sign-up Screen

Once verified, the user reaches the login screen, where they can enter their email and password to access the app. If the credentials are correct, they will be taken to the home screen. If they enter incorrect credentials, an error message will be displayed. Additionally, users have the option to reset their password, which will take them to the reset password screen.

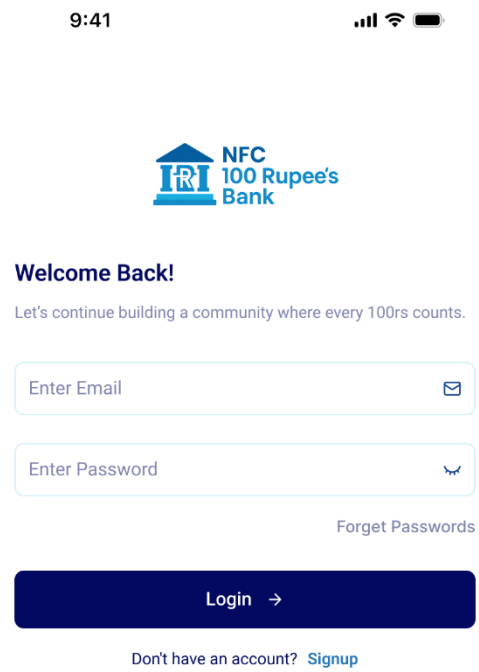


Figure 17: Log-in Screen

After login user will see home page. On the home page, users will find several features. At the top, there is a notification button, and to its right, the user's profile picture is displayed. Below that, there is a dynamic display showing the total amount donated,

which updates as they make donations.

The home page contains three main options: "Make a Donation", "Apply for Loan “and "Loan Repayment."

Make a Donation: When the user selects this option, they will be redirected to the payment gateway's checkout page. Here, they need to input the donation email, phone number, amount, select a bank, and complete the donation process. After finishing the donation, they will be redirected back to the home page.

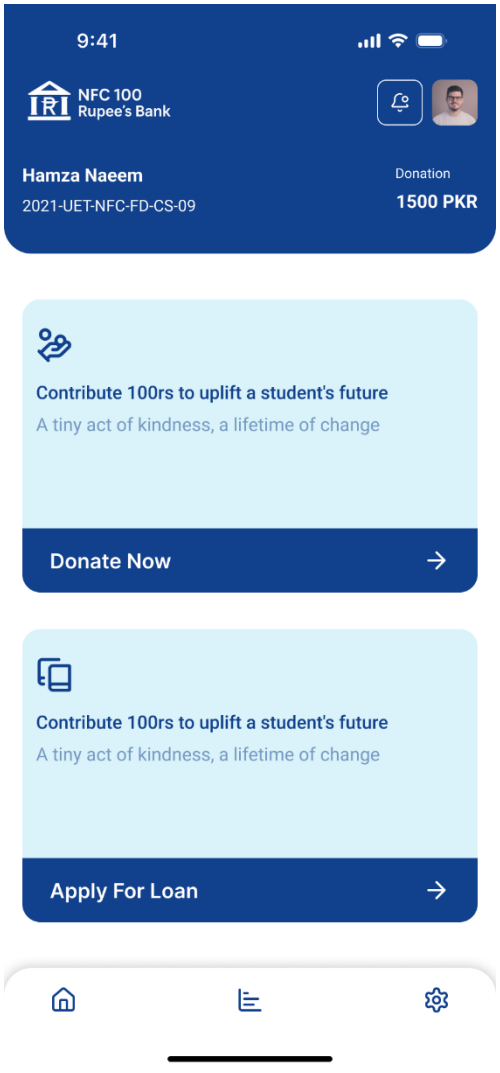


Figure 18: Home Page

Loan Repayment: If a user has taken a loan, then in this card his loan info shows in which he can see his monthly installment, remaining amount, next due date and his monthly status.

Apply for Loan: When users click on this option, they will be taken to the personal information screen, where they must fill in details such as registration ID, student name, current semester, email, CGPA, phone number, loan amount and upload front and back pictures of their CNIC.

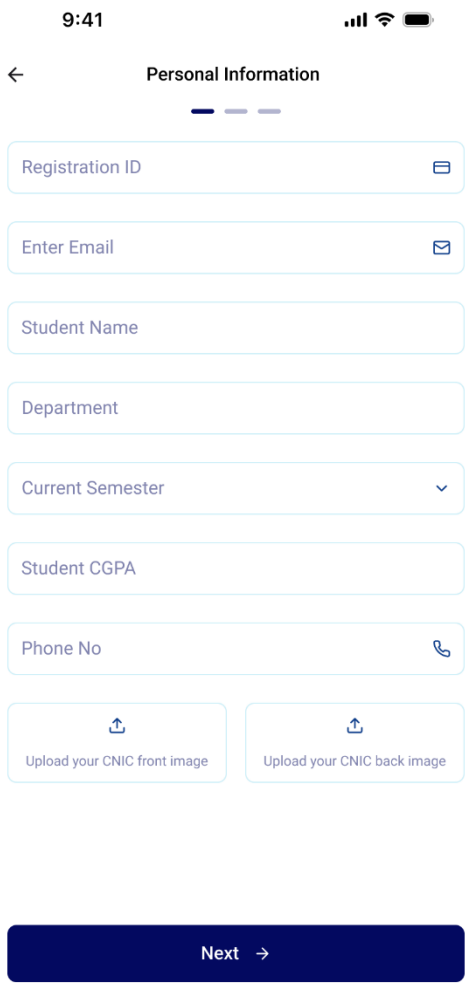


Figure 19: Personal Information (loan Application)

After completing this section, they will click "Next" to proceed to the family information screen, where they will provide details such as their father's name, father's occupation, father's salary, number of siblings, and father's phone number.

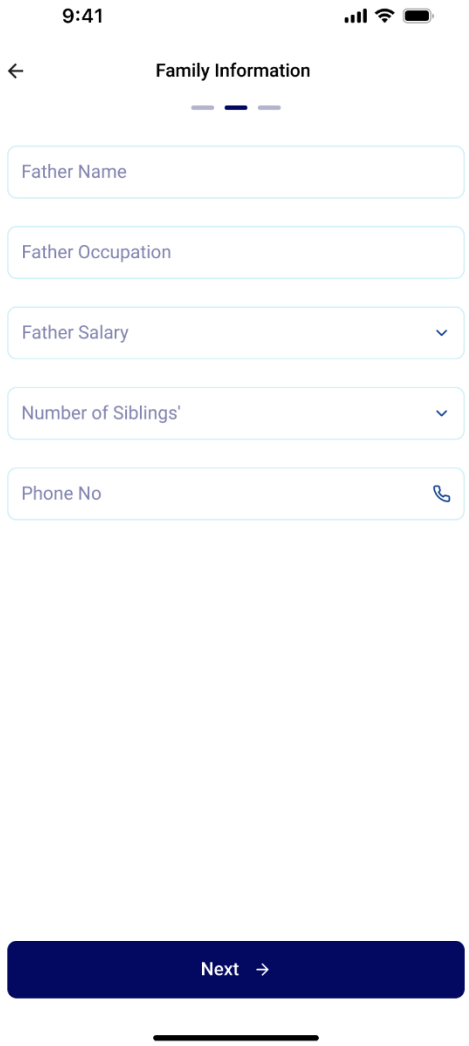


Figure 20: Family Information (Loan Application)

Once they fill out the family information, they will advance to the additional information screen, where they can upload pictures of their water, electricity, and gas bills. After submitting all the information, they can click on "Submit." This submission will be recorded in the database, and they will see a message confirming that their application has been submitted.

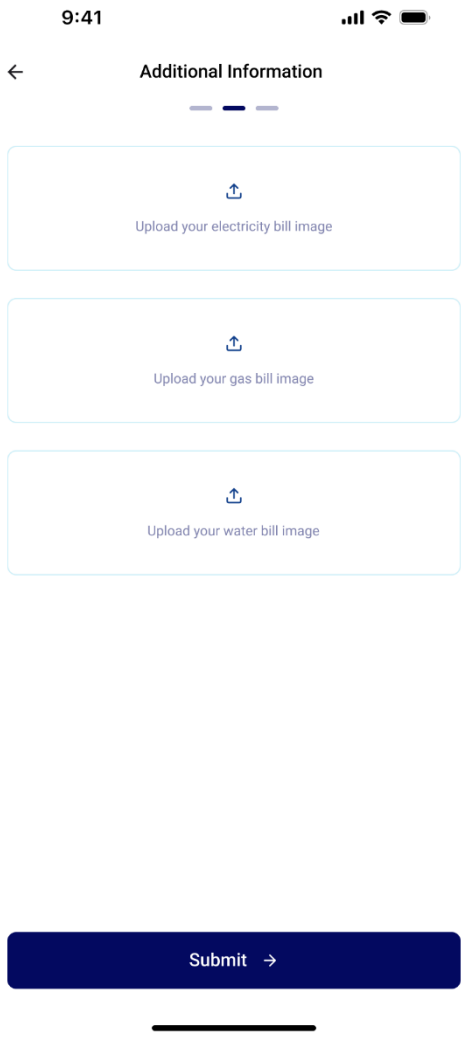


Figure 21: Additional Information (Loan Application)

At the bottom of the home page, there are three buttons: Home, Charts, and Settings. When the user clicks on the Charts button, they can view two types of charts: A donation over time chart, displaying monthly collection amounts in graph form. A category breakdown chart, showing which departments have received donations and the corresponding amounts.

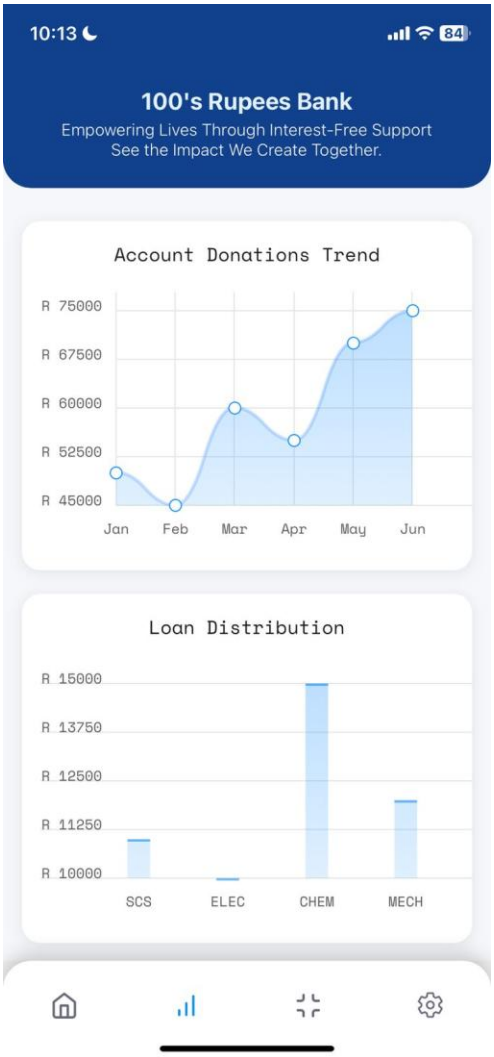


Figure 22: Chart Screen

Clicking on the Settings button allows users to access options to edit their profile, change their email, change their password, adjust notification preferences, view terms and conditions, learn about the app, and log out.

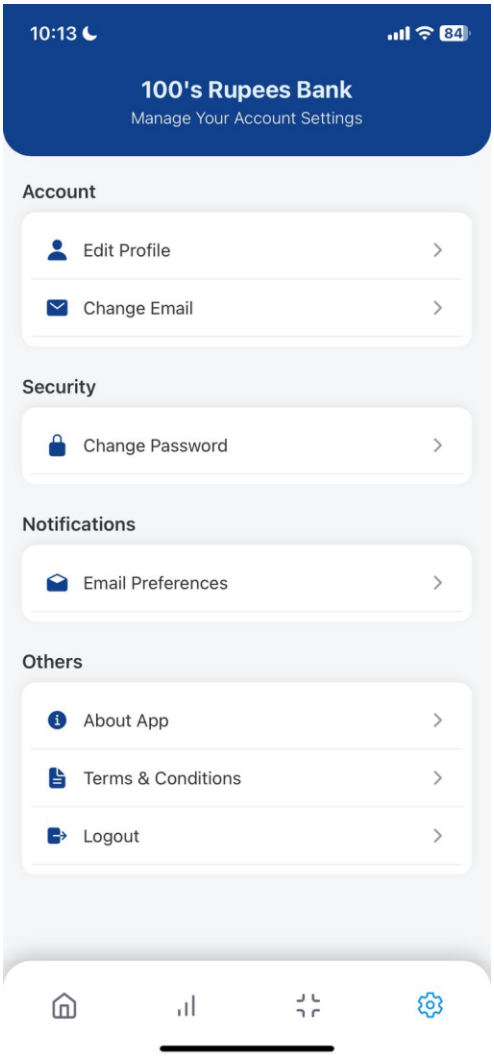


Figure 23: Setting Screen

4.3.2 Responsive Design Principles:

Responsive design principles were implemented to ensure that the user interface seamlessly adapts to different screen sizes and devices. The app is developed with a mobile-first approach, automatically adjusting to various screen sizes to provide a smooth and consistent experience across different devices. This responsiveness enhances usability and accessibility for students using different mobile phones.

4.3.3 Integration with Backend APIs:

The frontend is fully integrated with backend services via RESTful APIs. These APIs handle all critical interactions, such as user registration, authentication, donation processing, and loan application submissions. When users perform any action on the app—such as making a donation or submitting loan forms—the frontend sends the data to the backend server, which processes the request and updates the MongoDB database. This integration ensures real-time synchronization and a seamless user experience across the application.

4.4 Backend Development:

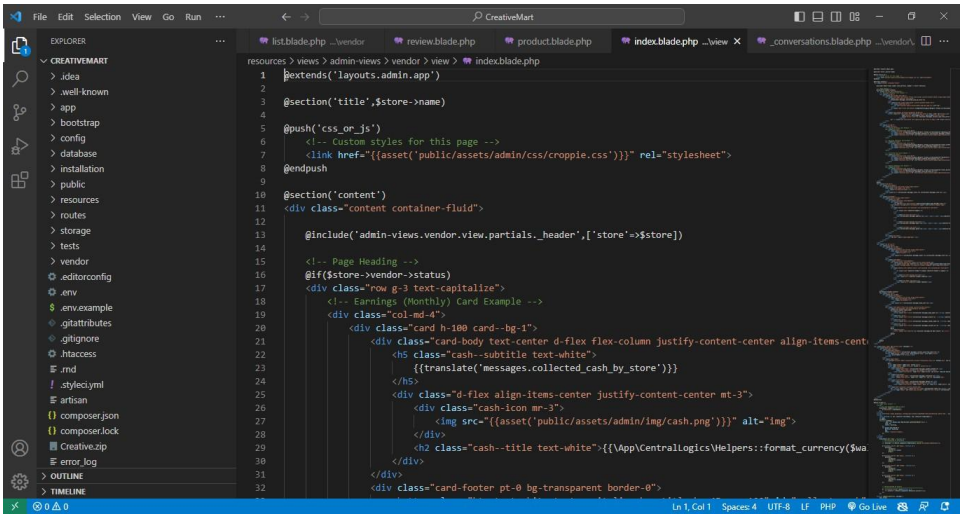


Figure 24: Backend Development

The backend handles the core functionalities of the system, including user management, donation handling, loan processing, and data security. Developed using a scalable architecture, the backend ensures efficient performance, secure data handling, and seamless communication with the frontend through API endpoints.

4.4.1 API Design and Implementation:

The backend was developed using Node.js with the Express.js framework, following RESTful principles to establish reliable communication between the frontend and the database. Each endpoint was created to serve a specific function—such as submitting forms, managing donations, or retrieving loan data—ensuring the system is modular and easy to maintain.

4.4.2 User Authentication and Authorization:

A secure, token-based authentication system (JWT) was implemented to protect sensitive student and admin data. During login, valid credentials generate a secure token that grants access to authorized resources. Passwords are hashed before storage to prevent breaches, and all user sessions are managed using encrypted tokens, ensuring the platform remains safe and private.

4.4.3 Donation and Loan Management:

The backend includes functionality for storing and managing donation records and loan application data. When a donation is made, details are recorded in the MongoDB database and reflected on the student's dashboard. Similarly, loan applications are submitted via the frontend, validated on the backend, and stored for admin review. Upon review, the admin's decision updates the loan status in the database, ensuring transparency and real-time feedback to the applicant.

Chapter 5: System Testing and Validation

5.1 Testing Methodology:

5.1.1 Types of Testing:

A structured and multi-layered testing methodology is followed to ensure the reliability, security, and usability of the NFC 100's Rupee Student's Bank application. The testing process included functional testing, performance testing, security assessment, and user acceptance testing to validate the system's readiness for deployment in an academic environment.

5.1.2 Test Scenarios and Use Cases:

Custom test scenarios were designed based on real-world user flows such as student registration, OTP verification, donation submission, loan application, admin login, and loan approval workflows. Each scenario reflected critical system functions and edge cases to validate system behavior under different conditions. These scenarios played a key role in verifying platform integrity and user experience.

5.2 Functional Testing:

Functional testing was conducted on both the mobile app and admin panel to ensure each module operated according to requirements. Tests confirmed the reliability of features such as login authentication, data entry validation in loan forms, successful donation payments, chart updates, and admin-side CRUD operations. Particular attention was given to form transitions and database updates after each user interaction to verify seamless integration across the stack.

5.3 Performance Testing:

5.3.1 Load Testing:

Load testing involved simulating a concurrent user environment to observe system responsiveness during peak usage, such as multiple users applying for loans or submitting donations simultaneously. Results indicated that the system could efficiently handle multiple active sessions without delay or crashes, given its lightweight Node.js and MongoDB-based architecture.

5.3.2 Stress Testing:

Stress testing pushed the application to its limits by simulating excessive requests, unexpected user inputs, and large file uploads (e.g., utility bill images). This helped reveal system behavior under strain and confirmed the platform's ability to recover from temporary failures like database timeouts or broken API calls.

5.4 Security Testing:

5.4.1 Vulnerability Assessment:

Security testing focused on identifying risks in authentication, form handling, and data transmission. Tools like Postman and Insomnia were used for manual testing of exposed endpoints, ensuring that unauthorized data access or role escalation was not possible. Measures were taken to secure sensitive routes and sanitize user inputs to prevent injection attacks

5.4.2 Encryption and Data Protection:

Sensitive information such as passwords was encrypted using hashing and secure storage techniques. Token-based authentication was implemented using JWT to protect session data. HTTPS protocols were considered for deployment to ensure secure communication.

5.5 User Acceptance Testing:

5.5.1 Involvement of Beta Testers:

A selected group of students and teachers were invited as beta testers to use the platform in a simulated environment. Their feedback on usability, layout clarity, and system responsiveness was recorded during interactions with the donation and loan modules.

5.5.2 Feedback Collection and Analysis:

Testers provided suggestions regarding form field arrangement, confirmation messages, and notification clarity. These insights were incorporated into the final app release through minor UI improvements and better form validation prompts, enhancing the overall user experience before rollout.

Chapter 6: Results and Discussion

6.1 Evaluation of Objectives

6.1.1 Student Experience Assessment:

The development of the NFC 100's Rupee Student's Bank aimed to provide a seamless and student-centric platform that offers both financial support and transparency. During testing and pilot usage, students found the mobile application easy to navigate, with a smooth loan application and donation flow. The onboarding, authentication, and dashboard layout were particularly well-received by students who appreciated the simplicity of viewing donation stats and accessing loan services without complicated steps. Real-time tracking of transactions helped reinforce confidence in the platform, proving that transparency and accessibility could be achieved even within a student-led ecosystem.

6.1.2 Community Engagement:

A core goal of the project is to encourage collective contribution and mutual support among students. Despite initial skepticism, the platform succeeded in cultivating a community-driven environment. Students will gradually began participating in the monthly contribution model once they were assured of oversight by NFC's Trust Fund. This shift indicated that with the right transparency mechanisms and trusted administration, students are willing to contribute toward the welfare of their peers. The system also ensured that students in financial need were able to apply for and receive timely, interest-free loans, reducing academic stress and potential dropouts.

6.2 Challenges and Mitigations:

The journey from concept to working prototype presented several critical challenges, especially in technical execution and community engagement. Each challenge required tailored solutions, balancing innovation with practicality.

6.2.1 Technical Challenges:

A major technical hurdle is accessing a Pakistani payment gateway's API credentials. Despite contacting several services like JazzCash, PayFast, and EasyPaisa, all providers required business documentation such as business owner Information, NTN, address verification, monthly payment volume and commercial registration even for sandbox testing. Since the project is student-led and non-commercial, these requirements could not be fulfilled. As a workaround, Stripe's test environment is used to simulate donation and repayment flows. This allowed frontend and backend modules to be tested and demonstrated successfully.

6.2.2 User Adoption Challenges:

One of the key adoption challenges is convincing students to contribute regularly to the welfare fund. Initially, many students will reluctant to donate due to concerns over how the funds would be managed. To build trust, it was decided that all operations would be supervised by a dedicated nonprofit organization the NFC Trust Fund to be formed and led by the university chairman. This approach provided the project with a strong foundation of accountability. Once this governance plan will share with students, trust in the system increased, and participation improved as students felt confident that their contributions were being overseen with transparency and institutional backing.

6.3 Lessons Learned:

The project's journey provided valuable insights and lessons that will inform future initiatives and improvements:

6.3.1 Iterative Development and User Feedback:

User feedback cycles helped the team continuously improve platform usability. Several UI and UX adjustments including clearer form fields, error prompts, and chart visibility were made in response to real user input. This iterative development process proved critical in creating a product that truly resonated with users.

6.3.2 Importance of Trust and Governance:

The experience clearly demonstrated that technical functionality alone is not enough user trust must be earned through transparency and governance. The suggestion to form the NFC Trust Fund was a direct result of this realization. A transparent financial aid system needs to be supported by credible administration, especially when contributions and disbursements are involved.

6.3.3 Payment Gateway Access Limitations:

Accessing FinTech infrastructure in Pakistan remains a challenge for student-led and non-commercial projects. The requirement of full business verification for even test APIs restricts innovation at the academic level. Future solutions could involve academic institutions forming partnerships with payment providers or supporting student developers in navigating compliance procedures.

6.3.4 Potential for Replication and Institutional Adoption:

This project has laid the foundation for a scalable and replicable model of interest free student lending. With proper oversight similar platforms could be deployed at other universities, potentially transforming student financial support across the country.

Chapter 7: Conclusions and Future Directions

7.1 Summary of Findings:

The NFC 100's Rupee Student's Bank project set out to address a critical challenge faced by university students access to ethical, interest-free financial assistance. By leveraging mobile technology and community-based funding, the platform successfully introduced a peer-supported digital solution that empowers students to contribute to and benefit from a collective financial ecosystem. The system provides a seamless and transparent user experience for both donors and loan applicants, ensuring that students in need can access funds without the burden of traditional interest-based lending.

7.2 Achievement of Objectives:

All primary objectives outlined at the inception of this project were met. The platform fosters financial inclusion by enabling interest-free loans, promotes a culture of student-led mutual aid, and maintains transparency in all financial activities through real-time tracking features. Additionally, the system encourages responsible participation and provides administrative control to ensure fair and accountable loan disbursement. The introduction of donation analytics and intuitive loan tracking tools has enhanced trust and usability among the student community.

7.3 Challenges and Lessons Learned:

Throughout the development and implementation phases, several challenges surfaced. One major limitation was the inability to access local payment gateway APIs without verified business credentials, which even restricted access to their sandbox environments. As a result, Stripe's test gateway was used to simulate payment workflows during development. Another significant challenge involved building student trust in the platform's fund management. This led to the proposal of establishing a formal nonprofit oversight body the NFC Trust Fund to be managed under the university's leadership. This suggestion was positively received by students and is expected to serve as a crucial enabler of long-term trust and participation.

Key lessons include the importance of institutional support in user adoption, the need for scalable technology choices, and the role of continuous user feedback in improving usability. These insights will be instrumental in guiding the future phases of the project.

7.4 Future Enhancements and Upgrades:

To extend the impact and utility of the platform, several high-impact enhancements are planned for future development. These will expand the app's role from a financial aid platform into a complete student financial bank system.

7.4.1 Student Payment Features:

The platform will evolve to support direct digital payments for a variety of academic expenses. This includes tuition fees, registration dues, summer course payments, library fines, hostel fees, and participation costs for university events. Students will be able to handle all such transactions within the app, reducing reliance on in-person payment methods.

A centralized fee management dashboard will also be introduced, offering students real-time visibility into their payment statuses, upcoming deadlines, and fine notifications. This will streamline financial planning and reduce the risk of penalties due to missed payments.

7.4.2 BNPL Model for Campus Life:

The platform also envisions the integration of a BNPL (Buy Now, Pay Later) model to support on-campus purchases. Students will be able to make deferred payments for cafeteria meals, bookstore items, stationery, and official university merchandise. Spending limits will be introduced based on student contribution history or university-approved limits, ensuring responsible usage. This functionality will turn the platform into an all-in-one student wallet designed for both academic and everyday use.

7.4.3 LMS and Academic Record Integration:

To automate and validate student identity and academic status, the platform will integrate directly with the university's Learning Management System (LMS) and student record systems. This integration will streamline verification, reduce manual review burden, and enable real-time confirmation of enrollment and semester progress.

7.4.4 Automated Reminders and Smart Notifications:

The platform will include intelligent reminders for upcoming loan repayments, donation opportunities, and payment due dates. These smart notifications will use behavioral patterns to engage students more effectively, improving timely participation and

reducing defaults.

7.4.5 NFC Trust Fund Governance Framework:

A formal governance structure NFC Trust Fund is planned to oversee operations, ensure compliance, and manage growth. This body will consist of university administrators, student representatives, and finance advisors to maintain transparency, credibility, and continuous improvement.

7.5 Final Remarks:

In conclusion, the NFC 100's Rupee Student's Bank has laid the foundation for a scalable, ethical, and student-powered financial platform. It addresses real-world challenges of tuition affordability through a simple, transparent model built on collective responsibility. The system has proven effective at delivering its intended goals, and with planned enhancements such as fee payment support, BNPL functionality, and institutional governance via the proposed NFC Trust Fund, it holds significant potential to become a national model for student financial empowerment in Pakistan.

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